

What makes a robot a robot?

Before we build one — what are we actually building?

We build robots to go where we cannot.



Mars

No air, no rescue. Rovers have worked there for years with nobody nearby.



Inside a reactor

Radiation that would kill a person in minutes. A robot just goes in.



Inside a person

Surgical robots move steadier than any human hand, through a smaller cut.



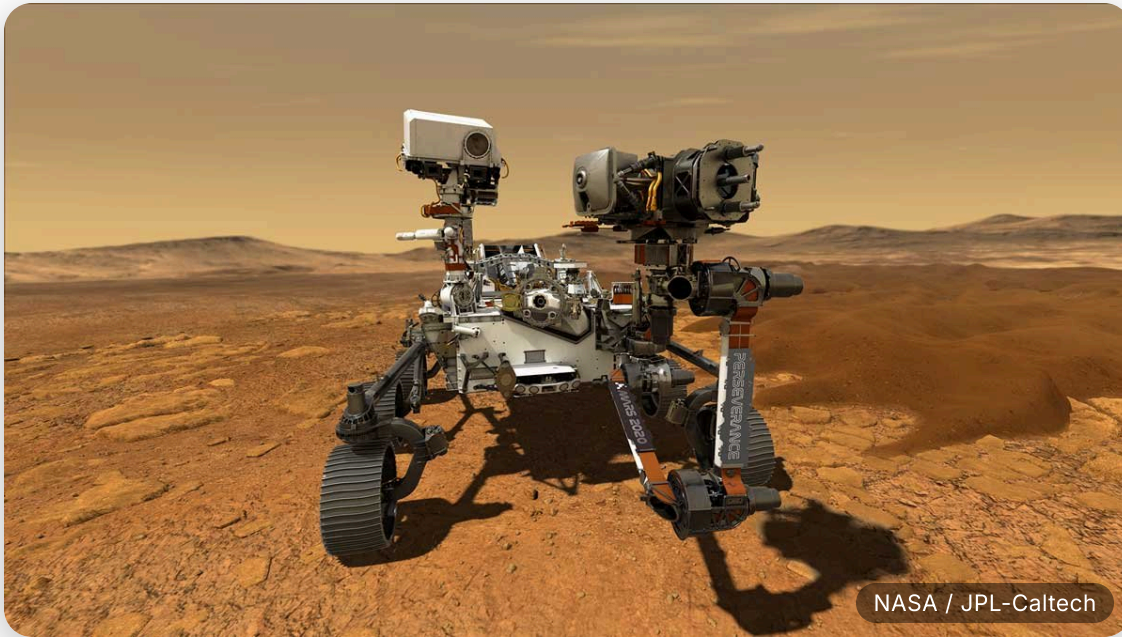
4 km down

Pressure that would crush a submarine. We map the seafloor this way.

Every one of them had to be **taught what to do** by somebody.

WHY THEY CANNOT JUST BE REMOTE CONTROLLED

Mars is too far away to drive.



3–22 min

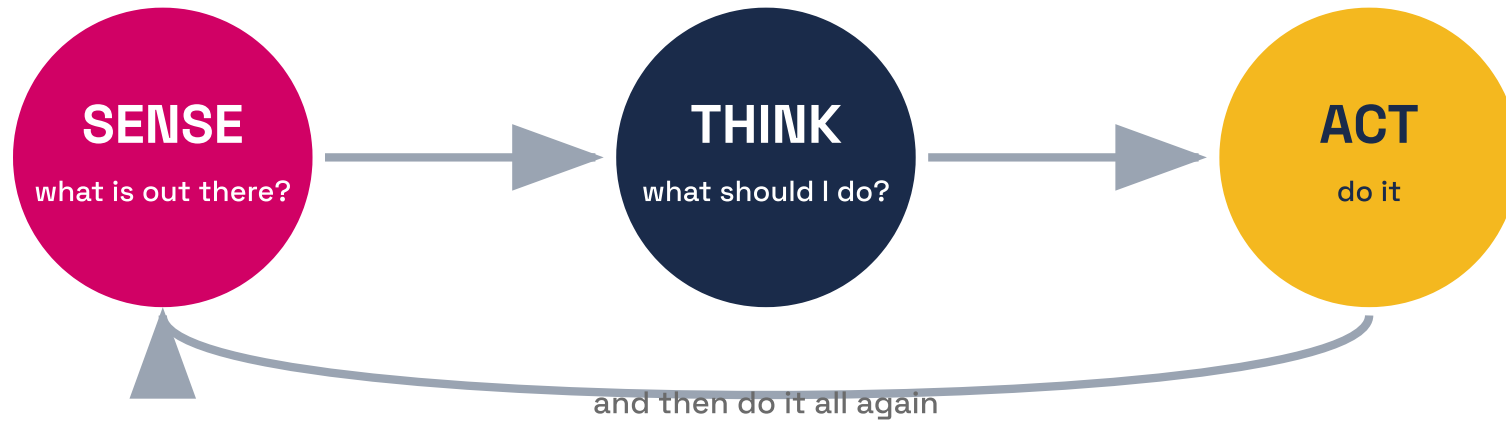
for a command to reach Mars — and just as long for the answer to come back

Do the maths. You see a cliff in a photo and send "STOP." That photo left Mars up to 22 minutes ago, and your command needs another 22 to get there. By the time the rover hears you, it drove off the cliff **three quarters of an hour** earlier.

So nobody drives it. **It has to decide for itself.**

WHICH MEANS IT NEEDS THREE THINGS

Sense. Think. Act. By itself.



That loop is the whole definition. **Miss any one and it is a machine, not a robot.**

Who is doing the thinking?

Toaster

NO

Acts on a timer. Senses nothing, decides nothing.

RC car

NO

Senses and acts — but **you** do the thinking.

Robot vacuum

YES

Finds a wall, decides to turn, drives on. Nobody watching.

Mars rover

YES

Has to be. Help is 22 minutes away.

A phone-controlled car is a *tool* — your brain is the missing piece, and you have to be there. **Take the thinking out of a person and put it inside the machine, and it becomes a robot.** That is the whole job.

The robot is not really the point.

You are going to be wrong a lot in here. On purpose. That is the part worth having.

- 1 Say what you think will happen** — before you find out. Out loud, in writing, where everyone can see it.
- 2 Test it properly.** Three runs, one thing changed at a time. Otherwise you learn nothing.
- 3 Be wrong without flinching.** A surprising result is the interesting one. It means something you believed was not true.
- 4 Find out why, and fix it.** Then defend your choice with the numbers — not with “it looked better.”

Those four work on **anything**. The LEGO is just what we practise them on.

You will build the loop one piece at a time.

ACT

Motors turn wheels. You will make a robot drive an exact distance, turn a real 90°, and choose gears for speed or pushing power.

Weeks 1 – 6

SENSE

Sensors arrive, one at a time. Your robot stops being blind — it starts noticing the world instead of just moving through it.

Weeks 7 – 9

THINK

You decide what the robot does with what it senses — and in what order. At the edge, back off. On contact, push.

Weeks 10 – 12

This is the order you **learn** them — not the order they **run**. By Week 10 all three run together, sense first, in one loop, many times a second.

QUESTION TWO

So what is it made of?

THE BRAIN

EV3 brick

A small computer. It holds your program and tells everything else what to do.

HOW IT ACTS

Motors

Turn electricity into movement. Everything your robot *does* in the ring, it does with these.

HOW IT SENSES

Sensors

Touch, colour and ultrasonic. They report numbers — they do not decide anything.

THE BODY

Beams, pins, gears

The frame. It has to hold together when another robot hits it.

THE ENERGY

Battery

A robot with a flat battery is a paperweight. Charge it, every week.

THE INVISIBLE PART

Your program

You cannot hold it or drop it. Without it, nothing above does anything at all.

Three sensors. Three different questions.

Touch

"Am I being pressed — yes or no?"

The simplest answer a robot can get. Perfect for knowing you have found your opponent.

Week 7 • port 1

Colour

"How bright is the floor?"

Pointed down, it tells the ring floor from the tape edge — so you can stop before you drive out.

Week 8 • port 3

Ultrasonic

"How far away is that?"

Sends out a sound and times the echo — like a bat. Gives you centimeters.

Week 9 • port 4

A sensor never tells you *what* something is. Ultrasonic says "30 cm away" — it cannot tell a robot from a wall. **Deciding what that means is your job.**

Letters move. Numbers sense.

LETTERS — MOTORS		NUMBERS — SENSORS	
A	Your first robot's motor (Weeks 1–2)	1	Touch (Week 7)
B	Left drive wheel (Week 4 on)	2	Spare
C	Right drive wheel (Week 4 on)	3	Colour, facing down (Week 8)
D	Spare	4	Ultrasonic, facing forward (Week 9)

Week 4 is the day this changes. Your drive motors move from A to B and C — so a program written in Week 1 will not drive the new robot. You rebuild it. That catches everyone; now it will not catch you.

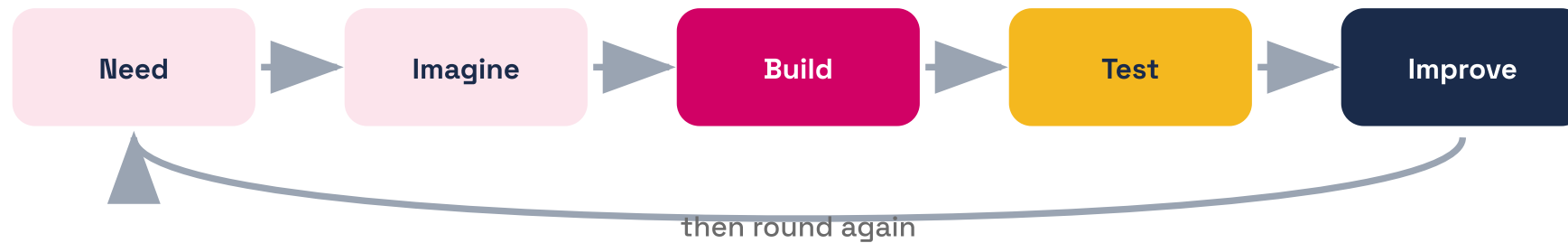
The robot does exactly what you wrote.

Not what you meant. Not what you wanted. Exactly what you wrote, in exactly the order you wrote it — every single time.

When your robot does something odd, it is not being difficult and it is not broken. It is following your instructions perfectly. **The bug is in the instructions** — and finding it is the actual skill.

This is why we **predict first**, then measure. The gap between the two is where the learning is.

Nobody gets it right first time.



A surprising result is a clue, not a failure. Your notebook is where you keep the evidence — predict first, keep the numbers, sketch what changed.

DECEMBER 9

Where this is going.

A sumo tournament, in front of your families. Two robots in a ring. Nobody touching the controls.

Your robot will have to **find** its opponent, **stay** inside the ring, and **decide** what to do when both happen at once — on its own, in about a second.

You will also be asked to explain *why* you built it that way. "We measured, and this was better" beats "it looked cool."

Next: open your notebook to **page 6** and build something that drives.

Build and drive

Tonight: Build Bevel Bot. Make it drive and stop.

0:00

Welcome

Find your team, check the kit, open the notebook.

5 min

0:05

Build

Bevel Bot from LEGO's pictures. Take turns. A mentor checks wheels, gears, pins and cables before anything runs.

25 min

0:30

Program & test

Your first program: motor A, drive, stop. Measure travel from the same front edge every time.

12 min

0:42

Evidence

Write the distance down, and one thing you noticed.

3 min

0:45

Clean up

Programs saved and named, kit counted, notebooks away.

5 min

0:50

Spare

Late starts, flat batteries, things that need rebuilding.

10 min

THE WORDS FOR TONIGHT

Electric motor — turns electrical energy into motion. **Program** — a set of instructions a computer follows. **Sequence** — the order those instructions run in.

PREDICT FIRST — BEFORE ANYTHING MOVES

Where will the first run stop?

DONE TONIGHT

Bevel Bot built, your own A-motor program, one measured drive and stop.

IF TIME REMAINS

Three 100 cm trials.

Find your grip

Tonight: Find the setup that slides at the steepest angle.

0:00

Arrive & set up

Ramp, angle gauge and catch area checked by a mentor.

5 min

0:05

Key idea & predict

Friction and bracing. Everyone writes a guess before the board is raised.

7 min

0:12

Build & test

Raise the board slowly. Record the angle at first slide, three times per setup. Change one thing only — the tyre or the surface.

25 min

0:37

Evidence & share

Work out mean and range, then compare with another team. Do you agree?

8 min

0:45

Pack up

Bevel Bots stay built — you need them again next week.

5 min

0:50

Spare

Resets, invalid trials, rebuilding.

10 min

THE WORDS FOR TONIGHT

Friction — the force that opposes sliding between two surfaces in contact. **Brace** — a support that helps a structure resist changing shape.

PREDICT FIRST — BEFORE ANYTHING MOVES

Which tyre or surface will slide at a steeper angle?

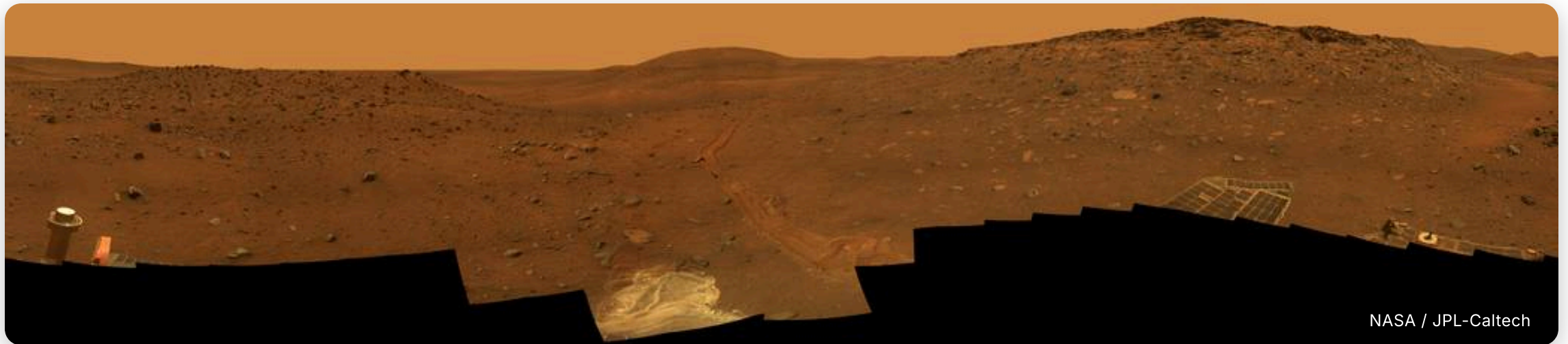
DONE TONIGHT

Two grip setups, three valid slides each, one brace comparison.

IF TIME REMAINS

Powered pushing demo · predict a third setup · does added weight change the angle?

NASA lost a rover to this.



APRIL 2009

Driving backwards, Spirit's wheels broke through a hard crust. Underneath was loose, slippery sand.

EVERY ATTEMPT

Each try at driving out dug it in deeper. The more they tried, the worse it got.

IT NEVER MOVED AGAIN

Built to last 90 days, it had already worked for five years. The mission ended here.

Not the motors. Not the cold. Not the computer. **It was grip.**

So they buried one on Earth.



A full-size copy of the rover, a tray of soil mixed to behave like the sand on Mars — and engineers deliberately driving it in to get stuck the same way.

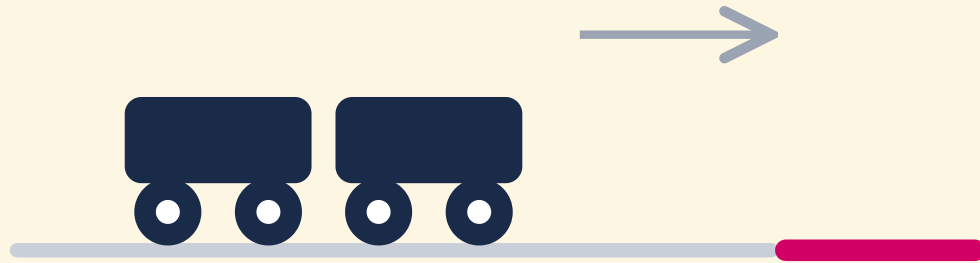
They tried escape moves here for **months** before sending a single command to Mars. Watching the wheels in this photo are Colette Lohr and Kim Lichtenberg, while Paolo Bellutta enters the next driving command.

Remember the 22 minutes? **You cannot take a command back** — so you test it somewhere you can.

The same thing decides December 9.

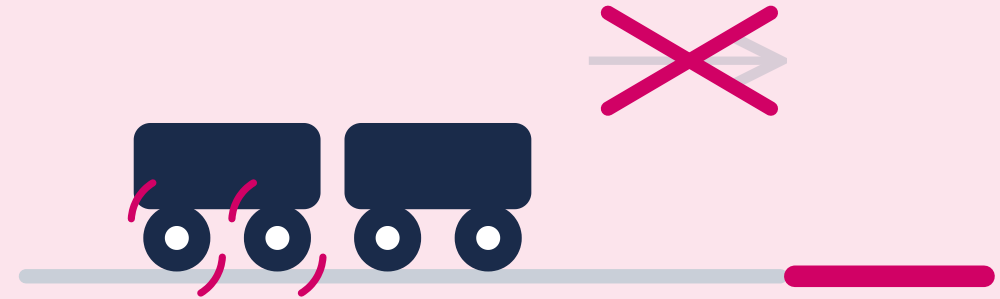
Tyres hold

Force reaches the floor. Your opponent goes backwards, towards the tape.



Tyres slip

Motors at full power, wheels spinning, nobody moving. All of it wasted.



THE THING WORTH REMEMBERING

Your motor does not decide how hard you can push. **Your tyres do.** Grip sets the ceiling — next week's gears only decide where you sit underneath it.

Three rules, or tonight measures nothing.

Change one thing

Same tyres when you compare surfaces. Same surface when you compare tyres. Change two and you will never know which one did it.

Mass • position • board all fixed

A roll is not a result

If it rolls or tips instead of sliding, the trial is **invalid**. Write it down, fix the setup, run it again — never average it in.

Reset flat every time

Three trials, then stop

Three tells you whether your setup is steady. A fourth tells you almost nothing. Spend the time on a harder question instead.

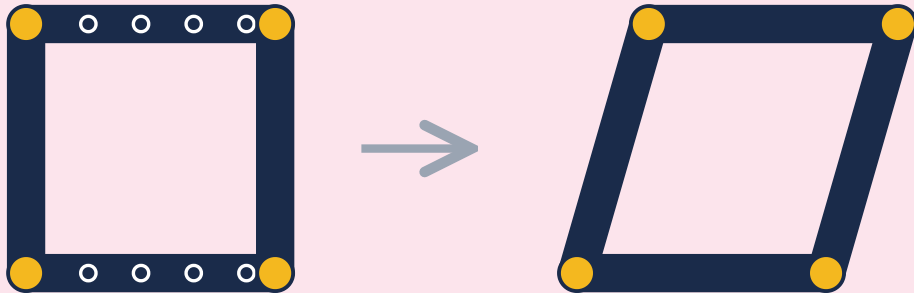
Then mean and range

Three trials are **clues, not proof** — but only if the setup stayed the same.

A rectangle is not rigid. A triangle is.

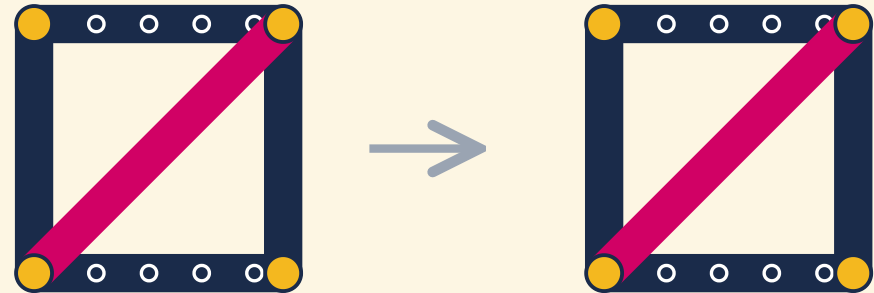
No brace

One pin per corner — and every pin is a hinge. The same push squashes it into a parallelogram.



One diagonal

Now it is two triangles. To squash it, the diagonal would have to stretch — and beams do not stretch.

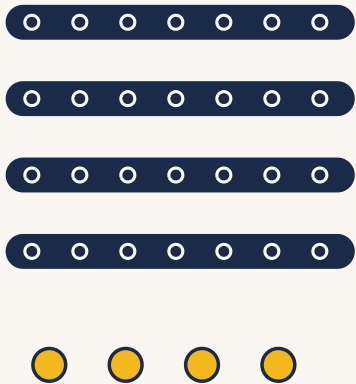


Nothing broke in the first one. **It just changed shape** — which is exactly what loses a sumo match.

Build one to pass around.

1 • THE PARTS

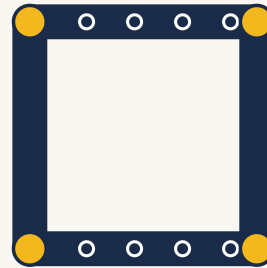
Four beams, four pins



All four beams the same length.

2 • PIN THE CORNERS

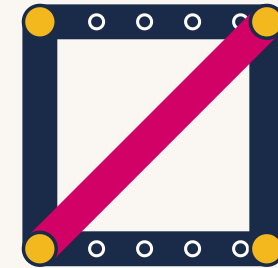
Push it — it squashes



Let them feel it give before you explain anything.

3 • ADD THE DIAGONAL

Same push — it holds



Then send them to find the same pattern on Bevel Bot.

- ▶ **Use the loose pins, not the friction ones.** Smooth pins pivot freely, so the squash is obvious.
- ▶ **Bigger racks better.** A small tight rectangle barely moves.
- ▶ **The diagonal must be longer than the sides** — about $1.4\times$. Check you have a long enough beam.
- ▶ **Pass it round first.** Feel it squash, then feel it refuse.

Three LEGO challenges. One of them fits.

BEST FIT

Robot Bulldozer

Builds a sturdy front blade, real contact with an opponent, and grip while pushing.

A small attachment on the robot they already have.

Cargo Carrier

Explores weight placement and secure mounting — both affect traction and balance.

Useful later. Building the tray takes time they do not have tonight.

Rescue Tow Truck

Explores traction under load and strong connections.

Less direct — towing links are not something sumo needs.

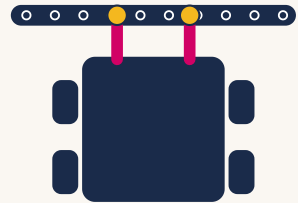
Bulldozer wins because the blade becomes a **sumo pusher** — and that is tonight's grip and bracing, used for something.

Build your sumo pusher.

Your robot can drive. **But can it move an opponent?**

Build

A horizontal beam across the front, clear of the floor and the wheels. **Two separated mounting points** — one will just pivot.



Test

Push a small, lightly loaded cardboard box along a marked 30 cm lane.

Observe

Do the wheels slip? Does the bar twist? Does the box slide sideways?

Improve

Change **one** feature — bar width, height, or bracing — and repeat.

SUCCESS LOOKS LIKE

The box crosses the line on three trials and the attachment never loosens. Same box, same surface, same motor setting. **Stop straight away if the motor stalls.**

IN THE NOTEBOOK

“What limited our pushing ability, and what evidence shows our change helped?”

Speed or shove?

Tonight: Compare gear setups. Pick your trade-off.

0:00

Arrive & set up

Lane marked, wall or board set for the distance sensor.

5 min

0:05

Key idea & predict

Gears, torque, gear ratio. Write your guess before anything turns.

5 min

0:10

Build the gear base

LEGO's gearbox model. Subassemblies are staged — take turns, and ask a mentor to check the mesh before you run it.

25 min

0:35

Swap gears & time it

Three runs on one gear pair. Swap to the other. Three more. **Change nothing else.**

10 min

0:45

Evidence & share

Mean and range for each pair. Which was faster, and what did it cost?

5 min

0:50

Spare

Rebuilds, re-runs, gears that will not mesh.

10 min

THE WORDS FOR TONIGHT

Gear — a toothed wheel that meshes with another to pass on rotation.

Torque — the turning effect of a force about an axis. **Gear ratio** — driven teeth ÷ driving teeth.

PREDICT FIRST — BEFORE ANYTHING MOVES

Which will be fastest? Which will pull best?

DONE TONIGHT

Gear base built, two gear pairs tested, three runs each.

IF TIME REMAINS

A third gear pair · work out the speed · a mentor-led tug test.

Fast, or strong? You only get one.

Fast

Gets across the ring first. Reaches the edge before anyone can stop it. But it can be **shoved aside**.

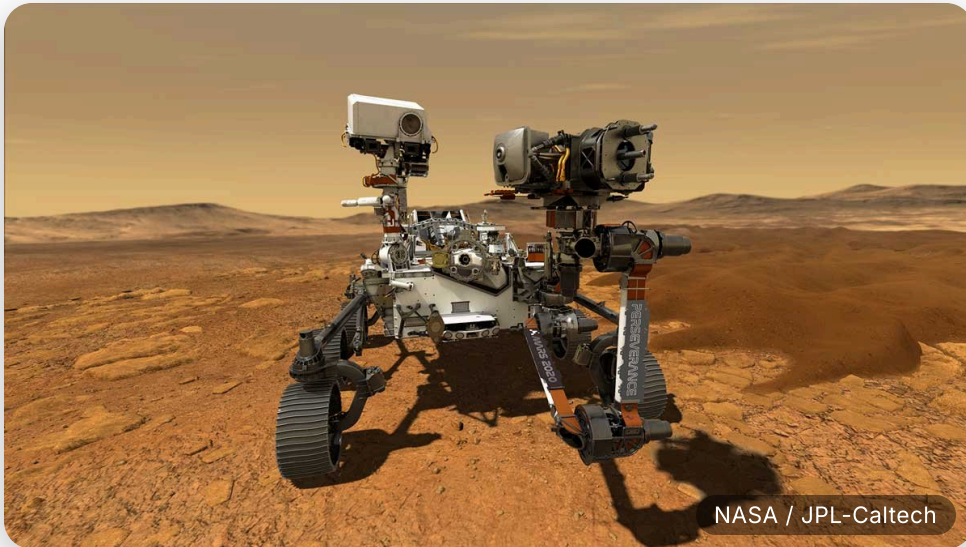
or

Strong

Pushes anything out of its way. Almost impossible to move. But it is **slow getting there**.

Your robot has **one motor**, and it always turns the same. So what decides which kind of robot you have? **Someone else already faced this exact choice.**

152 meters per hour.



That is the top speed of NASA's newest Mars rover. Not a typo — **a hundred and fifty-two meters in an hour.**

ABOUT 33× SLOWER THAN YOU WALK

~40 MINUTES TO CROSS A FOOTBALL FIELD

UNDER 200 WATTS — A CAR USES 150,000

Engineers who can land on Mars built something slower than walking. **Why would they do that?**

Because it can climb a rock as tall as your knee.

40 cm

The height of rock it will drive straight over. It clears obstacles as big as its own wheel — 52.5 cm.

Knee-high, to you

45°

The tilt it is built to survive without going over. Drivers keep it under 30° anyway.

Your ramp, last week

6 motors

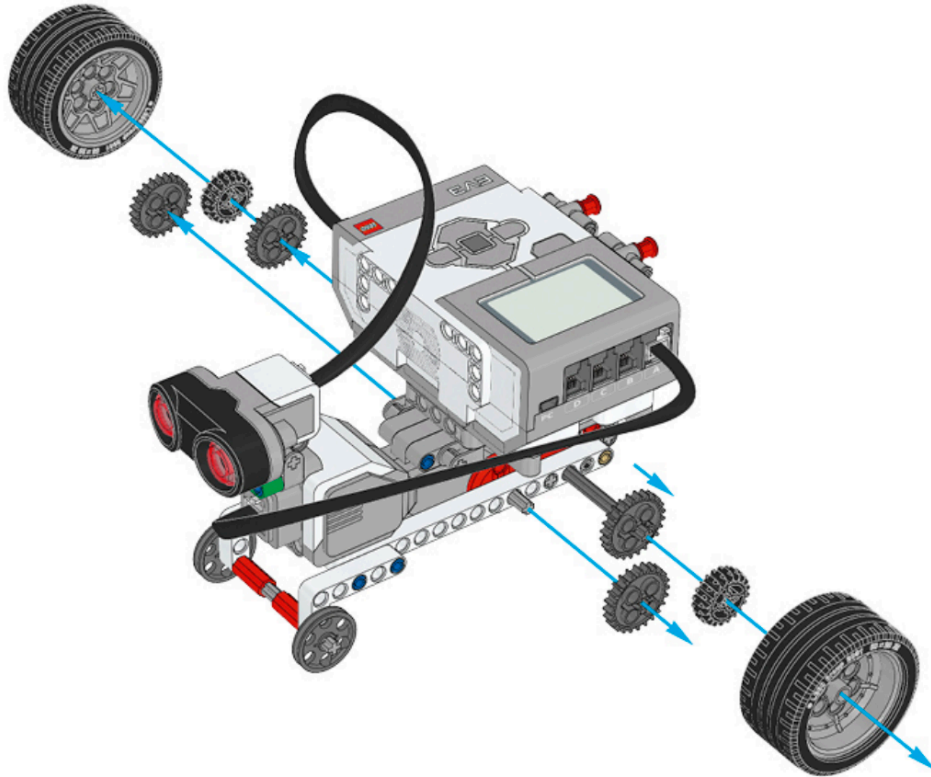
One per wheel, plus four more just for steering. All of it geared for pulling, none of it for racing.

Torque over speed

TONIGHT'S TRADE

NASA gave up speed and bought climbing with it. **You cannot have both** — gears let you choose which one you get, and tonight you measure the cost.

The gears come off. That is the point.



One motor. A gear train to each wheel. The gears slide off the axles — so you can **put different ones on and run it again.**

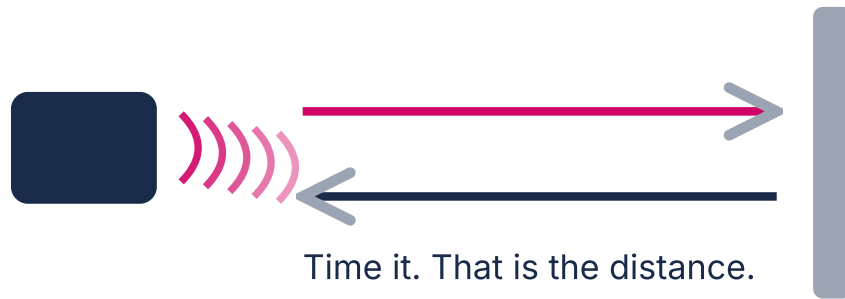
Everything else stays bolted together. Same brick, same motor, same wheels, same weight. **The gears are the only thing that changes** — which is what makes your two numbers worth comparing.

Building instructions © LEGO® Education • MINDSTORMS EV3 Core Set
45544

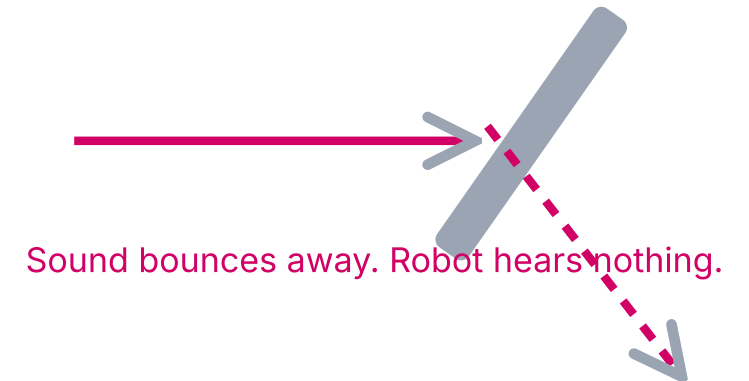
Build it once, then change one thing. **But how will you measure what changed?**

You don't. The robot measures itself.

Flat on: the echo comes back



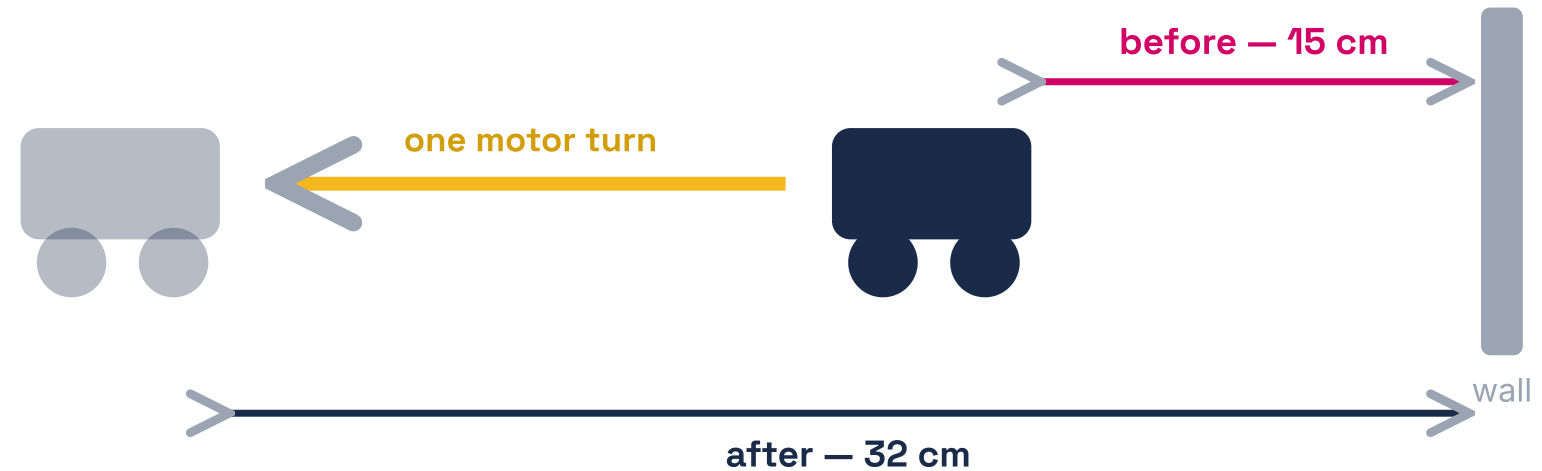
At an angle: it never comes back



WHY THAT MATTERS

That black unit with two round eyes on the front of your gear base shouts a sound at the wall and **times how long the echo takes to come back**. No tape measure, no chasing it across the floor. **Tonight it is your ruler** — and in Week 9 the same sensor becomes how your robot finds its opponent.

Press one button. The robot does the rest.



1 • PARK IT

At least **10 cm** from the wall, sensor square on to it.

2 • PRESS

The centre button. It reads the wall and **remembers** the number.

3 • ONE TURN

The motor turns **exactly once**, then stops. Same every run.

4 • READ IT

It measures again. The **difference** is how far you moved.

Speed is distance $\div$ time. That is all it is.

- 1 The formula you already know

$$\text{speed} = \text{distance} \div \text{time}$$

- 2 The sensor gives centimeters. The motor gives degrees.

Centimeters are fine as they are — or $\div 100$ for **meters**, the unit scientists use.

Degrees need $\div 360$ to become **whole turns**.

- 3 Three numbers, one formula

SPEED • CM/S

cm $\div$ seconds

SAME SPEED • M/S

(cm $\div$ 100) $\div$ s

MOTOR • RPS

(deg $\div$ 360) $\div$ s

TRY IT — 30 CM IN 4 SECONDS

$30 \div 4 = 7.5 \text{ cm/s}$ • $0.3 \div 4 = 0.075 \text{ m/s}$ — the same speed, written two ways.

So — which will you build?

You started tonight with a question you could only guess at. Now you have **numbers**.

You measured

How far one motor turn carries each gear pair, three times over, with nothing else changed.

You found the cost

Whatever you gained in one direction, you paid for in the other. That is the trade.

Now you choose

In Week 4 you build the robot you keep. This is the evidence you choose its gearing with.

REMEMBER WHAT NASA PICKED

They chose **slow** — 152 meters an hour — because on Mars, getting stuck is worse than arriving late. **Your ring is not Mars**. What does it reward?

Four things a distance reading can mean.

A plausible nearby reading

Something reflective is close. It **may be a wall**, not your opponent.

A far reading, or no echo

No confirmed target. **Do not turn that into a charge command.**

An error or disconnection

A detected fault. Stop moving and investigate.

A reading that looks fine but is wrong

The program cannot tell. This is the one that beats you — test angled targets, background objects and interference.

THE HABIT

A number on the screen is not the truth — it is **evidence about the truth**. Week 9 keeps the motors off entirely, so you can find out what the sensor really knows before you let it drive anything.